



2.12

Logarithmic Function Manipulation

Student Notes and Guided Practice

AP Precalculus - Unit 2 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Learning Targets

- Expand and condense logarithmic expressions.
- Apply product, quotient, and power properties with domain awareness.
- Use change of base to evaluate arbitrary-base logarithms.

Essential Question

How do logarithm properties translate multiplication and powers into addition and scalar multiplication?

$$\log_b(MN) \xleftrightarrow{\text{product}} \log_b M + \log_b N$$

$$\log_b(M/N) \xleftrightarrow{\text{quotient}} \log_b M - \log_b N$$

Concept Framework

Key Idea

Product property requires positive factors in the real logarithm setting.

Key Idea

Power property brings an exponent down as a multiplier.

Key Idea

Change of base allows evaluation using common or natural logs.

Key Idea

Algebraic simplification must preserve domain restrictions.

Formula Map**Formula and Structure**

$$\log_b(MN) = \log_b M + \log_b N$$

Formula and Structure

$$\log_b(M/N) = \log_b M - \log_b N$$

Formula and Structure

$$\log_b(M^p) = p \log_b M$$

Formula and Structure

$$\log_b x = \frac{\ln x}{\ln b}$$

Worked Examples**Worked Example 1**

Problem. Expand $\ln \frac{x^3 \sqrt{x-1}}{(x+2)^2}$.

Solution. $3 \ln x + \frac{1}{2} \ln(x-1) - 2 \ln(x+2)$, assuming all arguments are positive.

Worked Example 2

Problem. Condense $2 \log_5 x - \frac{1}{2} \log_5 y$.

Solution. $\log_5(x^2/\sqrt{y})$.

Worked Example 3

Problem. Evaluate $\log_2 7$ using natural logarithms.

Solution. $\ln 7 / \ln 2$.

Multiple-Representation Connection

AP Reasoning

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. Expand, condense, or change base as appropriate: $\ln(x^3y^2)$.

2. Expand, condense, or change base as appropriate: $\log_2 \frac{a^4}{b\sqrt{c}}$.

3. Expand, condense, or change base as appropriate: $2\ln x - 3\ln y$.

4. Expand, condense, or change base as appropriate: $\log_5 m + \log_5 n - \log_5 p$.

5. Expand, condense, or change base as appropriate: $\log_7 12$.

6. Expand $\ln \sqrt{\frac{x^2-1}{x^4}}$ with domain conditions.

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is _____.
2. A restriction I must remember is _____.
3. One graphical feature I can justify algebraically is _____.